## Short Answer

1. Give two important characteristics of physisorption
2. Differentiate physisorption and chemisorption
3. In case of chemisorption, why adsorption first increases and then decreases with temperature?
4. Which will be adsorbed more readily on the surface of charcoal and why? $NH_3$ or $O_2$ ?
5. Heat of adsorption is greater for chemisorptions than physisorption. Why?
6. Peptising agent is added to convert precipitate into colloidal solution. Explain with an example.
7. What happens when a colloidal sol of $Fe(OH)_3$ and $As_2S_3$ are mixed?
8. What is the difference between a sol and a gel?
9. Why are lyophillic colloidal sols are more stable than lyophobic colloidal sol.
10. Addition of Alum purifies water. Why?
11. What are the factors which influence the adsorption of a gas on a solid?
12. What are enzymes? Write a brief note on the mechanism of enzyme catalysis.
13. What do you mean by activity and selectivity of catalyst?
14. Describe some feature of catalysis by Zeolites.
15. Give three uses of emulsions.
16. Why does bleeding stop by rubbing moist alum
17. Why is desorption important for a substance to act as good catalyst?
18. Comment on the statement: Colloid is not a substance but it is a state of substance.
19. Explain any one method for coagulation
20. Write a note on electro osmosis
21. Write a note on catalytic poison
22. Explain intermediate compound formation theory of catalysis with an example
23. What is the difference between homogenous and hetrogenous catalysis?
24. Describe adsorption theory of catalysis.



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